



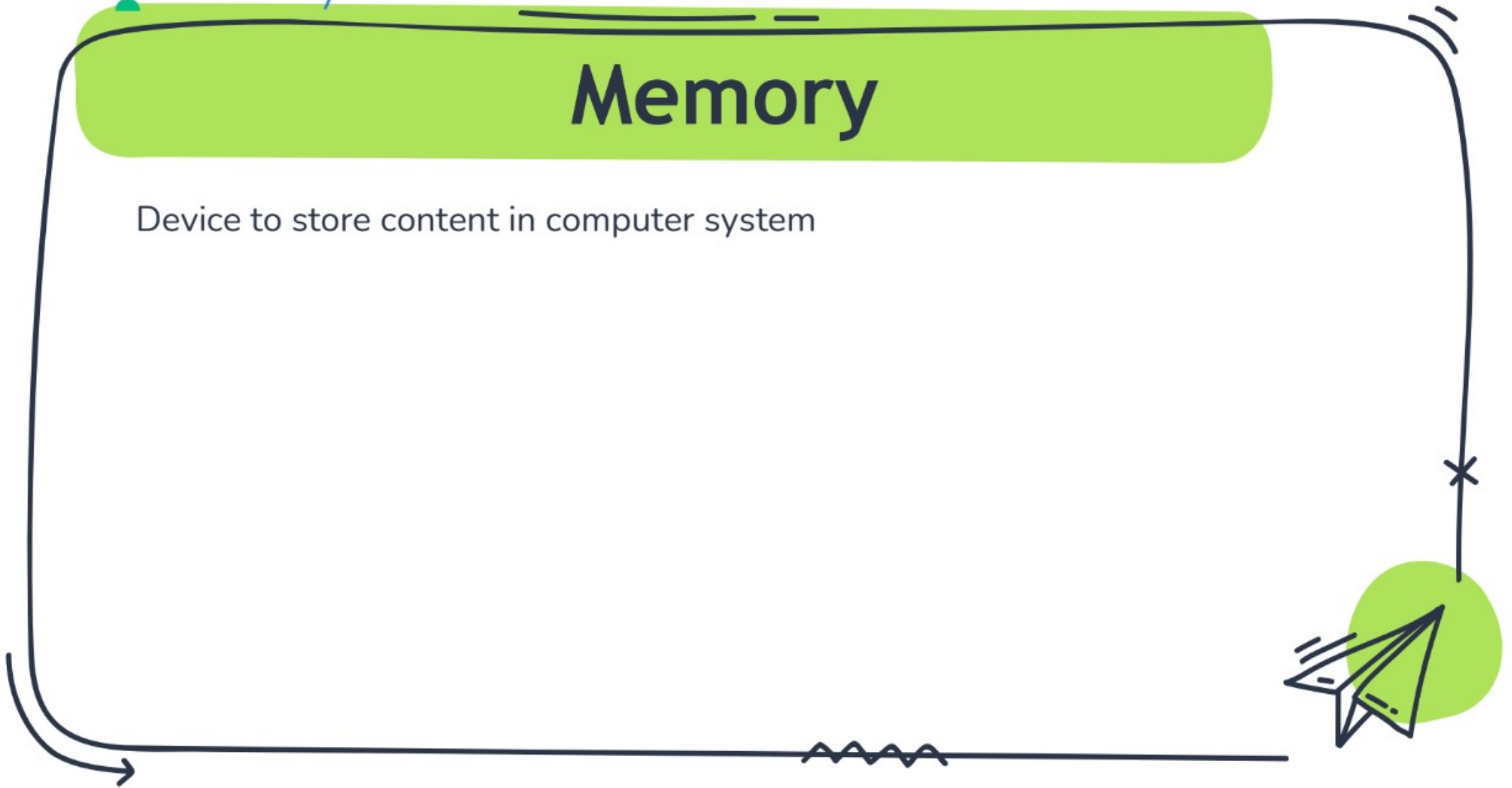
COA 2.0

MEMORY HIERARCHY & ADDRESSING

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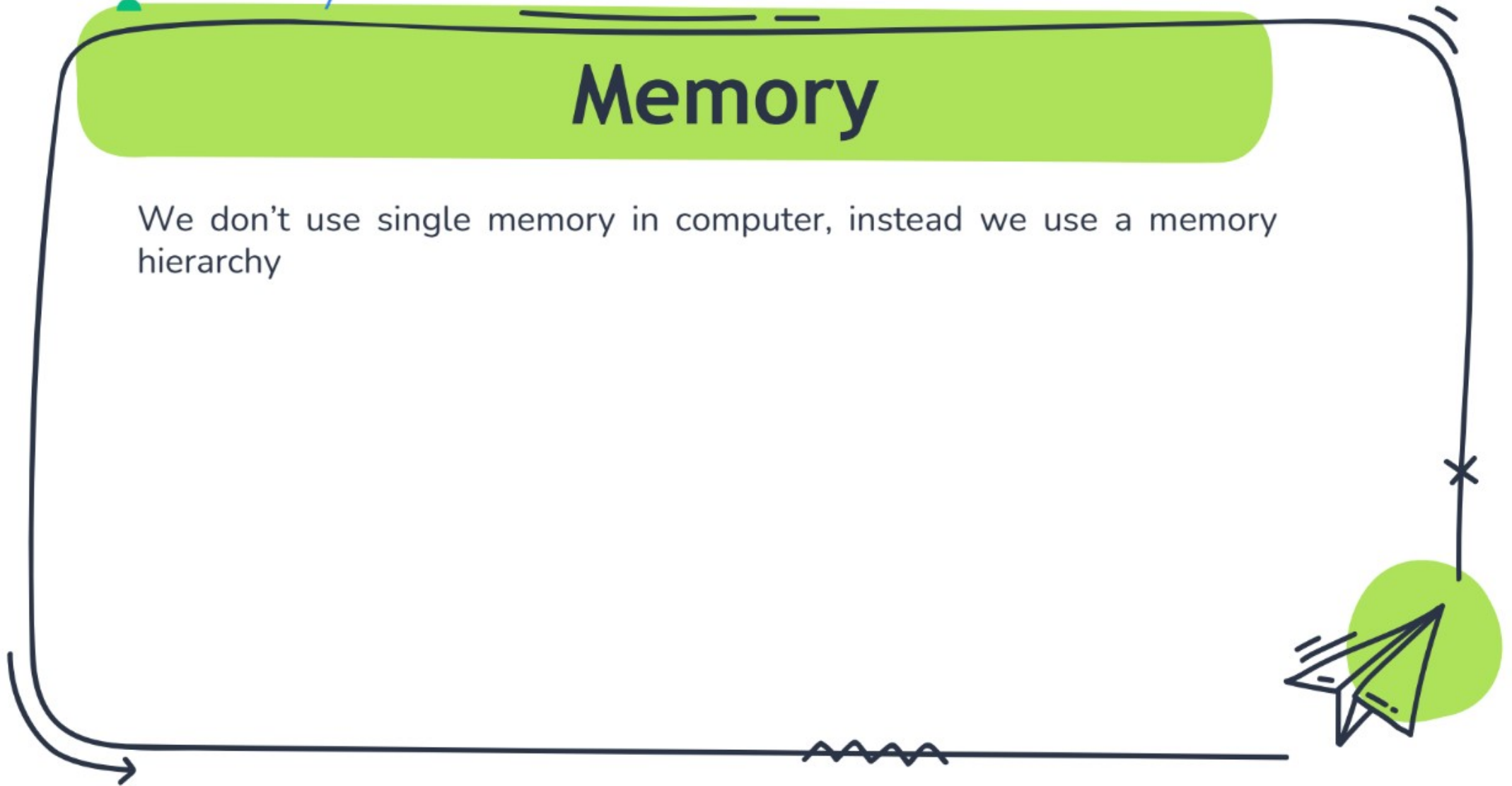
Memory

Device to store content in computer system



Memory

We don't use single memory in computer, instead we use a memory hierarchy



Memory Hierarchy

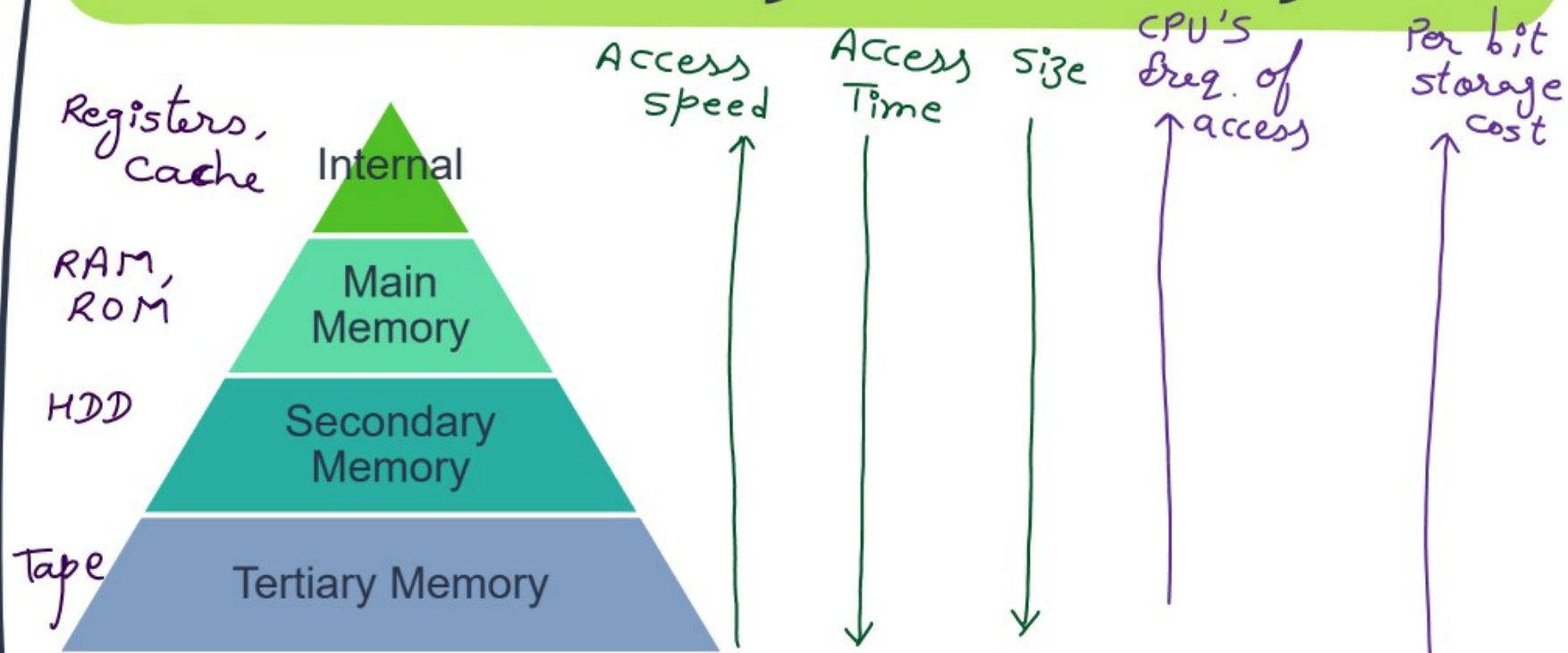
X Memory hierarchy used when discussing performance issues.

X **Goal of Memory Hierarchy:**

1. To maximize the Access Speed
2. To minimize the Per Bit Storage Cost



Memory Hierarchy



Average Cost of Memory

Memory	Size in Bits	Per Bit storage Cost	Cost of a Memory
M1	S1	C1	$S1 * C1$
M2	S2	C2	$S2 * C2$
M3	S3	C3	$S3 * C3$

Total memory hierarchy cost = $S1 C1 + S2 C2 + S3 C3$



Average Cost of Memory

$$= \frac{S_1 C_1 + S_2 C_2 + S_3 C_3}{S_1 + S_2 + S_3}$$

$$= \frac{\sum_{i=1}^n S_i C_i}{\sum_{i=1}^n S_i}$$

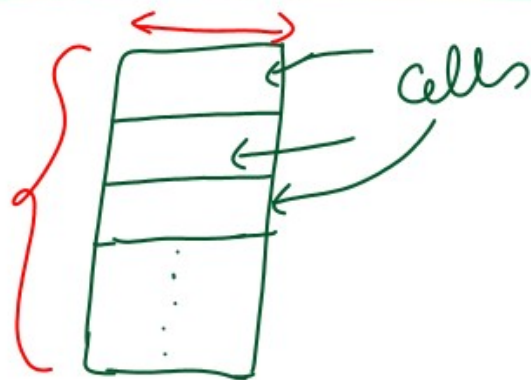


Memory Representation

Memory is represented as:-

= No. of cells $\times$ 1 cell capacity

= No. of memory locations $\times$ Bits per location



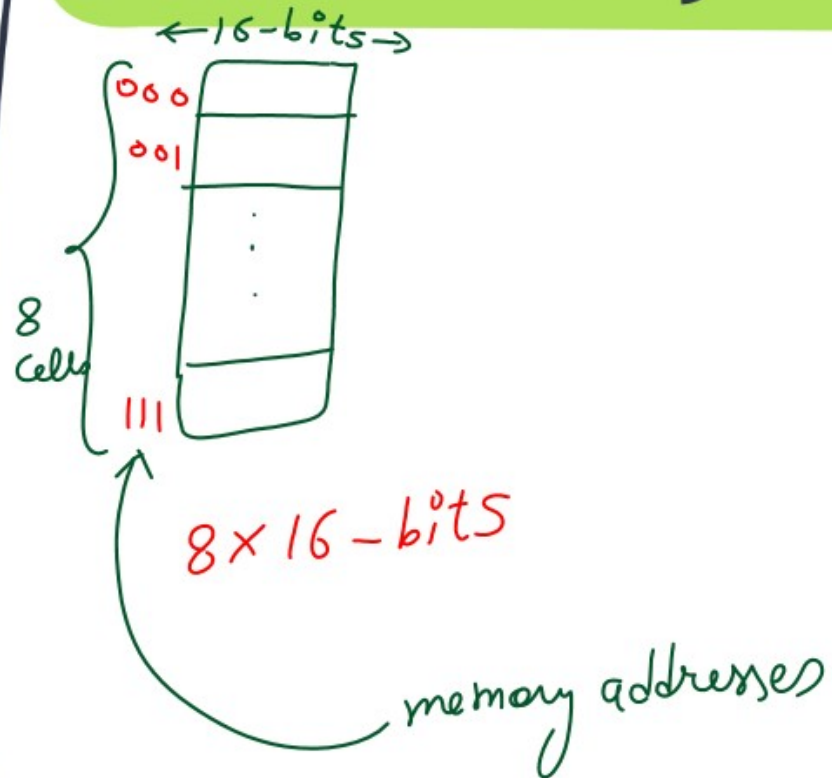
00	8-bits
01	8-bits
10	8-bits
11	8-bits

4 cells

4x8-bits



Memory Representation



Memory cycle time :-

Time required to perform
Read or write on one
cell in memory.



Memory Representation

No. of cells	4	8	16	2^n	∞
Address size	2-bits	3-bits	4-bits	n -bits	$\log_2 \infty$ - bits

Memory

word addressable

1 cell capacity = 1 word

(default)

Byte addressable

1 cell capacity = 1 byte



Byte Addressable Memory

Assume,

$128K \times 8\text{-bits}$

or

$128K \times 1\text{ Byte}$

or

$128K\text{ Byte}$

$b \Rightarrow \text{bits}$

$B \Rightarrow \text{bytes}$

Default
unit
of
storage = Bits

$$k = 2^{10}$$

$$M = 2^{20}$$

$$G = 2^{30}$$



Question 1

Memory is represented as?

- a) $A \times B$ where A = No. of memory locations, B = No. of bits in each location
- b) $2^a \times B$ where a = No. of address bits, B = No. of bits in each location
- c) $B \times A$ where, B = No. of bits in each location, A = No. of memory locations
- ☒ d) (a) & (b) both



Question 2

A memory has 16-bits address bus. Then how many memory locations are there?

- a) 64K
- b) 65536
- c) ~~216~~ 2^{16}
- d) All

$$\begin{aligned}\text{no. of memory locations} &= 2^{16} \\ &= 65536 \\ &= 64K\end{aligned}$$



Question 3

Ans = 16

Consider a CPU which has 13-bits address bus and 16-bits data bus.
Maximum size memory which can be supported by CPU is _____ Kbytes.

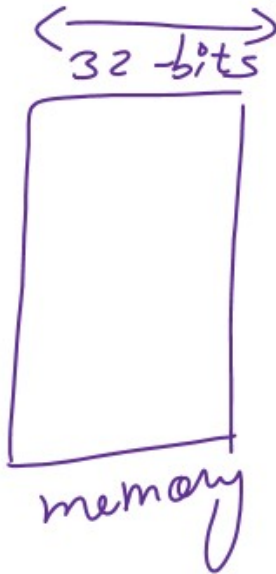
$$\text{Max memory} = 2^{13} \times 16\text{-bits} = 2^{13} \times 2\text{B} = 2^{14}\text{B} = 16\text{KB}$$



Ans = 64

Question 4

A 32-bits wide memory has 24-bits addresses to be accessed.
Maximum memory capacity is 64 MB?



$$\begin{aligned} 2^{24} \times 32\text{-bits} &= 2^{24} \times 4B \\ &= 2^{26} B \\ &= 64 MB \end{aligned}$$



Question 5

The memory cycle time of a memory is 200nsec. The maximum rate with which the memory can be accessed ____ **5 M** Bytes/Sec?

Note: Consider memory as byte addressable.

In 200 nsec time, mem. accessed for = 1 B

In 1 nsec —||—||— = $\frac{1 \text{ B}}{200 \text{ nsec}}$

In 1 sec —||—||—||— = $\frac{1 \text{ B}}{200 * 10^{-9} \text{ sec}}$

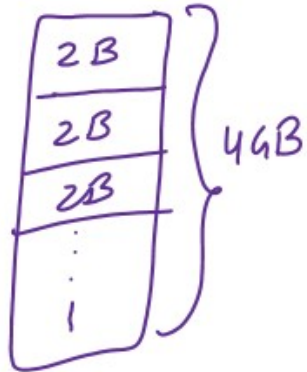
$$= \frac{10^9}{200} \text{ B/sec} = 5 \text{ MB/sec}$$



Question 6 GATE-2016

Ans = 31

A processor can support a maximum memory of 4 GB, where the memory is word-addressable (a word consists of two bytes). The size of the address bus of the processor is at least 31 bits?



$$\begin{aligned}\text{No. of cells in memory} &= \frac{4\text{GB}}{2\text{B}} \\ &= 2^6 \\ &= 2^{31}\end{aligned}$$

add. size = 31-bits



Question 7

Consider a memory of size 8GB. The number of ^{address} bits required for accessing the memory is _____ bits, if:

1. Memory is byte addressable $\Rightarrow 33\text{-bits}$
2. Memory is word addressable (word = 32 bits) $\rightarrow 4\text{Bytes} \Rightarrow \text{Ans} = 31$

$$1. \quad 8\text{GB} = 8\text{G} \times 1\text{B} \Rightarrow 2^{33} \times 1\text{B} \Rightarrow \text{add.} = 33\text{-bits}$$

$$2. \quad 8\text{GB} = 2\text{G} \times 4\text{B} = 2^{31} \times 4\text{B} \Rightarrow \text{add.} = 31\text{-bits}$$

$$\text{No. of cells} = \frac{8\text{GB}}{4\text{B}} = 2\text{G} = 2^{31}$$



Question 8

The address bus width of a memory of size 4096×8 bits is?

a) 10

b) 11

☒ c) 12

d) 13

$$2^{12} \times 8\text{-bits}$$

$$\text{Address} = 12\text{-bits}$$

